

25 August 2026

Editors

Journal of the American Statistical Association

Dear Editors,

We submit **“Surrogate-Assisted Local Conditional Distribution Inference with Scarce Reference Outcomes”** for consideration in the *Journal of the American Statistical Association*, Theory and Methods.

The paper studies the local conditional distribution

$$F_0(y \mid x_0) = \Pr(Y \leq y \mid X = x_0)$$

when a reference outcome is observed through selective two-phase verification, a lower-cost surrogate is broadly observed, and the verification fraction may vanish jointly with localization. This experiment arises in validation studies, sensor calibration, laboratory confirmation, expert review, and model-assisted evaluation.

The paper makes three connected contributions. First, it derives the observed-data canonical gradient and semi-parametric efficiency bound for a kernel-regularized conditional-CDF process, together with an exact positive-semidefinite covariance-excess identity for alternative augmentations under a known verification design. Second, it introduces one process-wide borrowing coefficient. The resulting quadratic variance criterion yields the transfer diagnostic $2G_h - H_h$, the path-oracle coefficient $\Pi_{[0,1]}(G_h/H_h)$, and a three-way estimator that is first-order equivalent to its fitted path oracle. Third, it establishes Gaussian-process, multiplier-bootstrap, and conditional-quantile inference when the reference fraction is fixed or vanishes, with effective local information $Mp_M h^d$. The scarce-reference covariance also yields a target-specific oracle square-root allocation benchmark.

The simulation study contains 119 configurations, including 24 settings for simultaneous bands. It covers transfer geometry, fitted-path adaptation, selective verification, scarcity, local reversal, estimated verification probabilities, non-Gaussian and discrete outcomes, and multivariate localization. Two real-data applications analyze EPA air-sensor and VitalDB pulse-oximetry data; a controlled synthetic AI-assisted review benchmark isolates selective review and local surrogate reversal. Together they span strongly aligned, weak, discrete, and locally reversed surrogate regimes.

The manuscript is original, is not under consideration elsewhere, and no manuscript or substantially overlapping version has previously been published, posted, presented, included in a thesis or dissertation, or deposited in a public manuscript repository. The Supplement contains complete proofs and extensions. The completed JASA Author Contributions Checklist, code, and derived numerical data sufficient to reproduce every reported table and figure accompany the submission.

Thank you for considering the manuscript.

Sincerely,

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